

Topic to discuss

Eulers Method

Numerical Problem

Homework Problem

YouTube @StartPracticing

Euler's Method :

Euler's method is a numerical technique used to find approximate solutions of a first-order differential equation.

It works by moving step by step using the slope
Euler's Method is also known as Runge Kutta Method of 1st order.

Standard Form :

$$\frac{dy}{dx} = f(x, y) \quad , \quad y(x_0) = y_0$$

$$\text{so, } y_{n+1} = y_n + h f(x_n, y_n)$$

where, h = step size

$$x_{n+1} = x_n + h$$

Numerical Problem:

Q: Solve $\frac{dy}{dx} = 1-y$, $y(0)=0$ using Euler's Method.

Find y , at $x=0.1$ and $x=0.2$

Compare the result with result of exact solution.

[Take $h=0.1$]

Solution: We have given the initial value,

$$x_0=0, \quad y_0=0, \quad h=0.1 \quad \text{and} \quad f(x,y)=1-y$$

According to Euler's Method, we have

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$\text{for } n=0, \quad y_1 = y_0 + h f(x_0, y_0)$$

$$\text{so, } y_1 = 0 + 0.1 f(x_0, y_0)$$

$$\Rightarrow y_1 = 0.1 \times (1 - y_0)$$

$$\Rightarrow y_1 = 0.1 \times (1 - 0)$$

$$\therefore y_1 = 0.1$$

$$\text{So, } x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$y_1 = y(x_1) = y(0.1) = 0.1$$

Now, Again applying Euler's Method formula,

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for, $n=1$

$$\begin{aligned}\Rightarrow y_2 &= y_1 + h f(x_1, y_1) \\ &= 0.1 + 0.1 (1 - y_1) \\ &= 0.1 + 0.1 (1 - 0.1) \\ &= 0.19\end{aligned}$$

We know,

$$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$$

$$y_2 = y(x_2) = y(0.2) = 0.19$$

Ans.

Solving to calculate Exact Solution,

$$\frac{dy}{dx} = 1-y$$

$$\Rightarrow \int \frac{dy}{1-y} = \int dx \quad \text{Integrating both side}$$

$$\Rightarrow -\ln(1-y) = x + C$$

$$\Rightarrow \ln(1-y) = -x + C$$

$$\Rightarrow 1-y = e^{-x+C}$$

$$\Rightarrow 1-y = e^{-x} \cdot e^C$$

$$\Rightarrow 1-y = C e^{-x} \quad \text{--- (1)}$$

Converting log to exponent

$$\left[\begin{array}{l} \text{using If } \ln A = B \\ \Rightarrow A = e^B \end{array} \right]$$

Applying initial condition,

$$y_0 = 0, x_0 = 0$$

$$1 - y = C e^{-x}$$

$$\Rightarrow 1 - 0 = C e^{-0}$$

$$\therefore C = 1$$

Now put in eqn. — (i)

$$\Rightarrow 1 - y = C e^{-x}$$

$$\Rightarrow 1 - y = e^{-x}$$

$$\therefore y = 1 - e^{-x}$$

Ans.

So,

$$y(0.1) = 1 - e^{-0.1} = 0.0951625$$

$$y(0.2) = 1 - e^{-0.2} = 0.181269$$

Comparing both the result,

Euler's Method

$$y(0.1) = 0.1$$

$$y(0.2) = 0.19$$

Exact Solution

$$y = 1 - e^{-x}$$

$$y(0.1) = 0.095163$$

$$y(0.2) = 0.181269$$

$$\text{Error for } y(0.1) = 0.004837$$

$$\text{Error for } y(0.2) = 0.008731$$

Ans.

Homework Problem:

Q: Solve the equation using Euler's Method

$$\frac{dy}{dx} = 3x^2 + 1 \quad \text{with } y(1) = 2$$

estimate $y(2)$ using $h = 0.5$

solution: Given,

$$y' = \frac{dy}{dx} = f(x, y) = 3x^2 + 1$$

$$h = 0.5$$

$$x_0 = 1$$

$$y_0 = 2$$

We have to calculate $y(2)$ means

$$y(2) = y(x_2)$$

$$x_1 = x_0 + h = 1 + 0.5 = 1.5$$

$$x_2 = x_1 + h = 1.5 + 0.5 = 2$$

It means first we need to find y_1 and then y_2
Applying Euler's Method formula,

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$\begin{aligned} \text{for } n=0 \quad \Rightarrow y_1 &= y_0 + h f(x_0, y_0) \\ &= 2 + 0.5 [3x_0^2 + 1] \\ &= 2 + 0.5 [3 \cdot 1^2 + 1] \\ \therefore y_1 &= 4 \end{aligned}$$

Now applying Euler's method again to calculate y_2 ,

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for, $n=1$

$$\begin{aligned} \Rightarrow y_2 &= y_1 + h f(x_1, y_1) \\ &= 4 + 0.5 (3x_1^2 + 1) \\ &= 4 + 0.5 (3 \times 1.5^2 + 1) \end{aligned}$$

$$y_2 = 7.875$$

$$\text{So, } y_2 = y \text{ at } x=2 = 7.875 \quad \underline{\text{Ans.}}$$

Connect with me



[Start Practicing](#)



[i._am._arfin](#)



[Arfin Parween](#)



[Arfin Parween \(\[Subscribe here too\]\(#\) \)](#)



[Arfin Parween](#)



[Arfin Parween](#)